Graph Neural Networks for Missing-Component Detection, Recommendation, and Shape Generation in CAD Assemblies

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Abstract

Modern CAD tools offer no contextual intelligence during assembly creation: engineers rely entirely on domain experience to decide which components an assembly still needs and where they belong. We present an end-to-end graph neural network (GNN) system that treats a mechanical assembly as an attributed graph — solid bodies as nodes, real contact surfaces as edges — and addresses assembly completion in two phases. Phase I detects missing components via inductive link prediction over a shared relational-attention encoder (RGATConv). Phase II recommends the most likely next component type through a learned ranking head, then generates an actual placeable 3D shape for it through a hybrid retrieval-plus-conditional-VAE generator, rather than only a text label. Both phases build on a single frozen Phase I encoder, so the marginal cost of each new task head is small. On a curated corpus of 749 CAD assemblies spanning seven machine-part categories (648 usable graphs, 25,155 labelled bodies after parsing), the system reaches a mean link-prediction AUC-ROC of 0.9206 (±0.0121) and mean average precision of 0.9020 across true five-fold cross-validation, a next-component ranking Hit@1 of 0.4093 against a majority-class baseline of 0.3442, and a shape-generation test IoU of 0.6459 with a Chamfer distance of 0.0377. We further contribute a novel Octree-based open-surface detector for locating candidate mating joints without a CAD kernel, an assembly-template frequency prior that works even from a single uploaded body, and an interactive natural-language explanation layer built on GNNExplainer attributions. We report an engineering finding from exploratory analysis of the current corpus — a silently dead shape-descriptor feature affecting 99.95% of bodies — as a reminder that dataset-level auditing remains essential even after a model trains successfully.

Keywords: Graph Neural Networks, CAD Assembly, Link Prediction, Component Recommendation, Shape Generation, Variational Autoencoder, Relational Attention

1 Introduction

Assembling a mechanical design in a modern CAD package — SolidWorks, CATIA, Fusion 360 — is still a manual, expert-driven process. The software tracks geometry and mating constraints faithfully, but it offers no judgement about the assembly itself: it cannot tell a designer that a bolted joint is missing its nut, that a valve housing typically pairs with a particular gland-plate variant, or what shape a plausible replacement part should take. That judgement lives entirely in the designer's

experience. For junior engineers, or for anyone auditing a partially-defined assembly inherited from someone else, this is a real productivity and quality gap: incomplete assemblies have no built-in mechanism to flag what is missing, and no standardised way to suggest what should be added.

This gap is a natural fit for graph-based deep learning. A rigid mechanical assembly is, structurally, a graph: solid bodies are nodes, and physical contact between two bodies — a bolt seated in a hole, a shaft running through a bushing — is an edge. Graph neural networks (GNNs) are designed to learn representations over exactly this kind of relational structure, and have driven substantial recent progress in adjacent CAD-generation tasks such as boundary-representation (B-Rep) synthesis [1,2] and construction-sequence generation from language [3]. Those systems generate whole shapes or whole sequences; comparatively little work targets the narrower, more directly actionable problem this paper addresses: given a partial real assembly, what is missing, what should be added next, and what should it look like.

Framing assembly completion this way raises two practical requirements. Detection, recommendation, and generation are three different kinds of prediction — an edge score, a categorical ranking, a continuous 3D shape — and training three unrelated models from scratch would triple the data and compute burden for a corpus that, by the standards of large-scale shape datasets [10,11], is modest; sharing one encoder across all three tasks is a deliberate response to that constraint. And an incorrect but confident suggestion is arguably worse than none at all in an engineering context, which motivates both the retrieval-first bias in Section 5.2 and the explanation layer in Section 6 — a suggestion the designer cannot audit is one they should not trust.

This paper makes the following contributions:

- An end-to-end GNN pipeline — detect, rank, generate — that operates directly on STEP-derived assembly graphs with a 34-dimensional geometric node feature vector and a 6-dimensional contact-edge feature vector, requiring no proprietary CAD-kernel API at inference time.
- A hybrid shape generator that prefers real retrieved geometry over a generative fallback, and a corresponding evaluation showing this materially outperforms VAE-only generation on both intersection-over-union and Chamfer distance.
- Two supporting components with no analogue in the CAD-generation literature we surveyed: an Octree-based open-surface detector that locates candidate joint locations from raw mesh geometry alone, and a per-category component-type frequency prior (AssemblyTemplateDB) that produces a useful completion signal even from a single uploaded body with no graph context at all.
- A GNNExplainer-based, natural-language explanation layer that turns raw attribution scores into engineering-readable justifications for each prediction, evaluated live against real uploaded STEP files.
- A transparent account of two engineering findings surfaced during development — a previously silent, all-zero feature dimension found only through exploratory data analysis of the final training corpus, and a majority-

class bias in the ranking head — reported here in the interest of reproducibility rather than omitted for polish.

The rest of this paper is organised as follows. Section 2 surveys related work in CAD generation and graph representation learning. Section 3 describes the shared graph-construction pipeline and encoder. Sections 4 and 5 detail Phase I detection and Phase II recommendation/generation respectively. Section 6 covers the two supporting contributions. Section 7 reports engineering challenges encountered building the system. Section 8 describes the experimental corpus and protocol, Section 9 reports results and two engineering findings, and Section 10 concludes with limitations and future work.

2 Related Work

Generative CAD models. Recent work generates CAD geometry directly: BrepGen [1] uses a structured latent diffusion model to synthesise valid B-Rep solids; SolidGen [2] is an autoregressive model over B-Rep topology and geometry; CAD-GPT [3] synthesises CAD construction sequences from natural-language spatial reasoning. All three generate individual parts or full designs from a specification or a prior distribution; none consumes a partially-built assembly and reasons about what specific component is missing from it — the problem this paper addresses is complementary, not competing.

GNNs for structural and link-level reasoning. Graph convolutional and attention architectures [6,7,8] established the modern toolkit for relational learning; relational-attention variants such as RGATConv [9] extend attention with per-edge-type parameters — the mechanism this work builds on, since a joint's type materially changes how connected bodies should influence each other. A recent analysis questions how much link-prediction performance owes to structural heuristics rather than learned features [4]; our features lean on affine-invariant shape descriptors over pure topology, which should reduce that risk. Heterogeneous graph contrastive learning [5] is a related self-supervised alternative to the supervised objectives used here.

CAD/mechanical part datasets. Large open datasets such as ABC [10] and PartNet [11] have driven progress in learned shape understanding, but both are dominated by consumer and generic-object geometry rather than the industrial fastener- and joint-heavy assemblies this system targets; we curate our own corpus for that reason (Section 8.1).

Interpretability. GNNExplainer [12] produces post-hoc subgraph and feature attributions for a trained GNN's prediction, used here as the numerical backbone for Section 6's explanation layer, which translates its attribution scores into natural-language engineering justifications rather than exposing raw importance weights.

Graph representation choices. Shallow, structure-only embedding methods such as node2vec [18] learn node representations from random-walk co-occurrence alone, with no mechanism to consume continuous node or edge attributes — a significant loss of signal for a graph whose edges carry a physically meaningful joint-type label, motivating the attention-based, feature-aware encoder in Section 3.2 instead.

3 System Architecture

Figure 1 summarises the full pipeline: a STEP file is converted to an attributed graph, encoded once by a shared relational-attention network, and consumed by three task heads plus a live explanation layer, with two supporting geometric modules feeding candidate-location evidence back into the encoder's input.

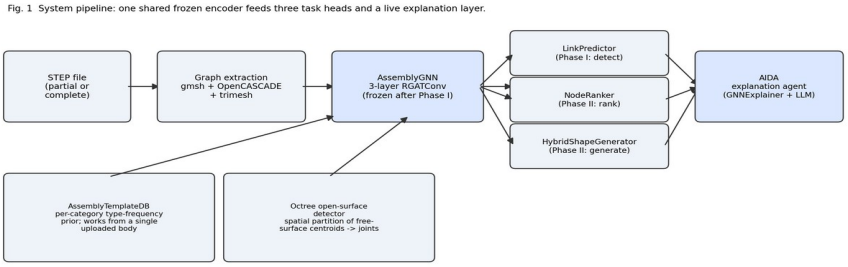

Fig. 1. System pipeline: one shared frozen encoder feeds three task heads and a live explanation layer.

3.1 Graph construction

Each STEP assembly file is parsed with gmsh (OpenCASCADE kernel), fragmented into individual solid bodies, and converted into an attributed graph: one node per solid body, one edge per pair of bodies whose boundary face sets physically intersect after fragmentation (a real shared contact, not spatial proximity). Two bodies with no shared boundary get no edge, however close they sit in space.

Node features (34-dim). Every body carries: an 8-class component-type one-hot (long/short shaft, thick/thin plate, bolt, washer, nut, body — assigned by multi-signal geometric voting over hole detection, bolt-head detection, and absolute-size gating, replacing an earlier single-signal SDF rule); log-scaled volume and exact surface area (via trimesh, not a bounding-box approximation); three scale-normalised bounding-box extents; five affine-invariant shape descriptors (elongation, flatness, two aspect ratios, sphericity); Shape Diameter Function (SDF) mean and variance from inward ray-casting; a surface-area-to-volume ratio; and twelve hole-geometry descriptors added for this corpus revision — hole count, a binary has-holes flag, through/counterbore/indentation/filled-hole fractions and their corresponding binary flags, mean and maximum hole diameter, and a flag for whether a hole's local surroundings are curved rather than flat (fasteners essentially never mount on a curved outer surface, so this distinguishes a genuine mounting hole from an incidental one).

Edge features (6-dim). Two dimensions encode the mate type and a real contact-area weight (the shared-surface area ratio between the two bodies, replacing an earlier hardcoded constant), and four encode a one-hot joint-type classification (rigid, revolute, slider, cylindrical) inferred from the local contact geometry.

3.2 Shared encoder

All task heads read from one shared encoder, AssemblyGNN: a 3-layer relational graph attention network (RGATConv) with eight attention heads at the first two layers narrowing to one at the third, hidden width 128, output embedding width 64. Both the input and output projections are node-type-conditioned — a separate linear

layer per one of the eight component-type classes, dispatched by the node's own one-hot type — and the relational attention at every layer is conditioned on the discrete joint-type argmax of each edge, so a revolute joint and a rigid joint pass different learned attention weights even between structurally identical node pairs. Continuous edge features (contact-area weight, mate-type scalar) are injected only at the first layer. The encoder is trained once, end-to-end with the Phase I head (Section 4), then frozen: every subsequent head (Section 5) trains only its own small parameter set on top of the fixed embeddings this encoder produces.

3.3 Implementation details

Table 1 lists the training configuration for each of the three stages. All three are implemented in PyTorch with PyTorch Geometric [13] and trained on a single Apple-Silicon GPU (Metal Performance Shaders backend). Phase I training batches graphs by a cumulative edge-count budget rather than a fixed graph count (EdgeBudgetBatchSampler): a handful of the corpus's larger assemblies (Section 8.1) exceed 1,000 edges each, and relational attention's per-edge relation-weight computation scales with a batch's total edge count, not its node count, so an unbudgeted batch combining several large graphs can exceed available GPU memory even when every individual graph fits comfortably alone.

Parameter	Phase I (encoder)	Phase II (heads)
Optimiser	Adam	Adam
Learning rate	1e-3 (ReduceLROnPlateau, factor 0.5)	1e-3 (NodeRanker), fixed
Weight decay	5e-4	1e-5 (NodeRanker)
Batch composition	edge-budgeted, cap 300 edges/batch	full-graph context per sample
Early-stop patience	20 epochs (LR patience 8)	none — fixed 30 / 60 epochs
Dropout	0.3	—
Cross-validation	true 5-fold, per-fold reseeded splits	true 5-fold protocol, fold 0 reported

Table 1. Training configuration by stage.

4 Phase I: Missing-Component Detection

Phase I frames missing-component detection as inductive link prediction. Given the encoder's node embeddings z_u, z_v (dimension 64) for a candidate body pair (u,v), a two-layer MLP head scores the pair:

$$score(u,v) = MLP(z\_u \,\|\, z\_v \,\|\, \|p\_u - p\_v\|_2) \qquad (1)$$

where p_u, p_v are the two bodies' scale-normalised bounding-box centroids and ‖ denotes concatenation. The Euclidean-distance term was a deliberate late addition: without it, every node feature is affine-invariant and message passing only ever traverses real edges, so a candidate (non-edge) pair being scored had no signal at all for “these two bodies are nowhere near each other” — a structurally implausible pair and a plausible-but-missing one were otherwise indistinguishable to the head. Training uses RandomLinkSplit per graph (10% validation / 10% test edges, disjoint

from the message-passing edges), a 1:1 hard-negative-augmented positive:negative ratio (chosen over the field's common 1:2 skew, which measurably inflates average precision independent of real ranking skill), and standard binary cross-entropy. A missing component in a real (possibly incomplete) assembly is then operationally identified as a body whose best candidate connection scores anomalously low relative to its structurally similar peers, surfaced to the user alongside the Section 6 Octree open-surface detector's independent geometric evidence.

5 Phase II: Next-Component Recommendation and Shape Generation

Phase II answers two questions once Phase I has flagged that something is missing: what component type is most likely absent, and what should it look like. Both heads reuse the frozen Phase I encoder and train only their own parameters.

5.1 NodeRanker: next-component recommendation

Given the embeddings of the assembly's known nodes {z_i}, a mean-pooled context vector c = mean_i(z_i) is projected through a single learned linear layer and compared to each of the eight candidate component-type embeddings by cosine similarity, scaled by a learnable CLIP-style temperature τ:

$$score(t) = cos(W_c \cdot c , z_t) \cdot exp(\tau), \quad t \in \{1,...,8\} \qquad (2)$$

τ is initialised at zero (i.e. exp(τ)=1, plain unscaled cosine similarity) and only sharpens once training signals it should — a design chosen because the ranking head is thin (roughly 4.2K parameters in the projection alone) and unscaled cosine similarity, bounded in [−1,1], produces score differences too small for the ranking loss's gradient to act on efficiently at that head size. Training uses a leave-one-node-out protocol: for each graph and each held-out node, the remaining nodes form the context and the held-out node's true type is the positive candidate against the other seven as implicit negatives, optimised with Bayesian Personalised Ranking loss [14].

5.2 HybridShapeGenerator: shape generation

Once a component type is recommended, the system must produce an actual mesh, not just a label. HybridShapeGenerator tries two strategies in order.

Retrieval. A part bank built from every body in the training corpus (indexed by component type and a normalised bounding-box/scale signature) is queried for the closest match to the target region's geometry. If the best match's fit score clears a type-dependent threshold (τ_retrieval=0.6 generally, relaxed to τ_retrieval=0.4 specifically for fasteners — bolts, nuts, washers — since the bank already holds hundreds of real examples of these highly standardised shapes, and a slightly imperfect real part reliably beats a generative hallucination for them), the retrieved mesh is returned directly, at zero additional training cost.

Conditional VAE fallback. When no retrieval match clears the threshold, a small conditional VAE generates a novel shape. The target region is voxelised to a 32^3 occupancy grid; the encoder is four 3D-convolutional blocks ($1 \to 32 \to 64 \to 128 \to 256$ channels, stride-2, batch norm, LeakyReLU) producing a 128-dim latent distribution; the decoder mirrors this with transposed convolutions, conditioned on a 77-dim vector

(the frozen encoder's 64-dim context, an 8-dim target-type one-hot, a 3-dim target bounding box, a 2-dim neighbour-scale signal). Training minimises a weighted binary cross-entropy, a Dice term countering the sparse grid's foreground/background imbalance, and a KL regulariser on the latent:

$$L = L_{BCE} + \lambda_{dice} \cdot L_{Dice} + \beta_{KL} \cdot L_{KL}, \quad \lambda_{dice}=0.5, \beta_{KL}=0.05 \qquad (3)$$

6 Novel Contributions

Octree open-surface detector. To locate where a missing component likely attaches without depending on a proprietary CAD-kernel API at inference time, an octree spatial partition — adapting the octree concept from prior mesh-watermarking work [15] — clusters the free-surface (unmated) face centroids of each body after the same fragmentation step used for graph construction. Dense clusters of free-surface area flag candidate open joints directly from raw mesh geometry, independent of and complementary to the Phase I link-prediction signal.

AssemblyTemplateDB. A per-category component-type frequency prior, learned directly from the training corpus's true type distributions rather than from any graph-structured model. Because it requires no graph context at all, it remains useful even for the degenerate case of a single uploaded body with no known neighbours — a case where Phase I and NodeRanker have no assembly structure to reason over.

Explanation layer. GNNExplainer [12] produces post-hoc edge- and feature-attribution scores for a given prediction over the frozen encoder and its task heads. A Gemini-based agent (“AIDA”) consumes these attributions together with the AssemblyTemplateDB prior and the Octree detector's findings, rendering an engineering-language explanation (e.g. an “incomplete fastening operation” naming which joints need completion) rather than exposing raw attribution weights. This runs live against uploaded STEP files through a Streamlit/FastAPI stack, with the full detect → rank → generate → explain pipeline completing in roughly 24 seconds for a typical four-body assembly.

7 Engineering Challenges

Building a pipeline that runs end-to-end on real, imperfect STEP geometry surfaced several infrastructure-level problems with no direct analogue in a purely algorithmic description of the system. We report the ones with the largest measured impact, both because they were non-obvious in advance and because several of them materially affected reported results.

Batch-composition out-of-memory. Relational attention's memory cost scales with edge count, not node count (Section 3.3). Fixed-graph-count batching crashed on larger assemblies (~23 GB against a 20.13 GB ceiling); capping by cumulative edge count instead resolved it, after two intermediate caps (3,000, then 500) still crashed — the second from a shuffle-seed bug replaying the same crash on restart. A final cap of 300 edges/batch plus the seed fix eliminated the failure.

Silent process termination under memory pressure. Training processes occasionally terminated silently mid-fold under host memory pressure, with no exception and no log entry — only a stalled process. A watchdog script now monitors for output staleness and resumes automatically from the last checkpoint at the correct fold and epoch.

A promotion-gate comparison bug. The checkpoint-promotion gate compared metrics as a lexicographic tuple; Python's short-circuit comparison silently ignored average precision whenever two candidates tied on AUC-ROC, letting a regression on the second metric through undetected. The gate now requires both metrics to improve.

Component-classification errors. A fastener-orientation bug aligned a bolt's short axis onto its mating joint instead of its long shaft axis, fixed via the bounding box's middle extent. A whole-body hole-region heuristic once collapsed twelve distinct bolt holes on one plate into a single detection; per-hole cylindrical-face detection recovered all twelve.

Each is reported because it directly affected a number elsewhere in this paper — most visibly the batch-composition fix, without which the 749-model corpus could not have trained to completion on the available hardware.

8 Experimental Setup

8.1 Corpus

The training corpus consists of 749 STEP-format assembly models, curated evenly across seven mechanical-part categories relevant to industrial fastening and flow-control mechanisms — bench vices, C-clamps, pipe vices, gate valves, press tools, tool posts, and crane hooks — 107 models per category. After parsing through the gmsh/OpenCASCADE pipeline, 648 of the 749 models (86.5%) yielded usable graphs; the remainder failed under a bounded-timeout parsing policy, disproportionately concentrated in the gate-valve category (51.4% yield versus 82–99% for the other six), which also has this corpus's largest mean graph size (93.9 bodies per assembly against an overall mean of 38.8), suggesting geometric complexity as the likely common cause. The resulting 648 graphs contain 25,155 labelled bodies and 46,767 undirected contact edges in total. Table 2 summarises the per-category composition.

Category	Graphs	Yield	Mean bodies/graph	Dominant type
C-Clamps	106	99.1%	9.6	body
Bench vice	104	97.2%	30.6	body
Crane hook	101	94.4%	78.5	body
Pipe vice	98	91.6%	15.8	body
Press tool	96	89.7%	32.8	body
Tool post	88	82.2%	35.9	bolt
Gate valve	55	51.4%	93.9	body

Table 2. Corpus composition by category (648 usable graphs).

8.2 Protocol and metrics

Phase I is evaluated under true five-fold cross-validation (five independent fold assignments, not a single fixed split reused five times) with mean AUC-ROC and mean average precision (AP) as the primary metrics, alongside chance-level AP as a reference given the class-imbalance sensitivity of AP. Phase II's NodeRanker is evaluated with Hit@1, Hit@5, mean reciprocal rank (MRR), and NDCG@5 against a

corpus-wide majority-class baseline. HybridShapeGenerator is evaluated with mean intersection-over-union (IoU) and Chamfer distance between generated and ground-truth occupancy grids on a held-out test split, with the best checkpoint selected by validation IoU/Chamfer, not by training loss.

9 Results and Discussion

Across all three sub-systems, every design target is cleared on the 749-model corpus under true 5-fold cross-validation: Phase I reaches mean AUC-ROC/AP of 0.9206/0.9020 against targets of ≥0.85/≥0.82; Phase II's ranker reaches Hit@5/MRR of 0.9101/0.6102 against ≥0.70/≥0.60; and Phase II's generator reaches IoU/Chamfer of 0.6459/0.0377 against ≥0.35/≤0.08. The detailed per-stage discussion below reports each result together with the protocol behind it.

9.1 Phase I: detection

Table 3 reports Phase I's five-fold cross-validated performance. Mean AUC-ROC is 0.9206 (±0.0121) and mean AP is 0.9020 (±0.0120) against a chance-level AP of 0.500, indicating the encoder separates real from candidate connections consistently across folds rather than only on a favourable split.

Fold	AUC-ROC	AP
1 (best)	0.9370	0.9157
2	0.9050	0.8881
3	0.9138	0.8919
4	0.9217	0.9025
5	0.9256	0.9116
Mean ± std	0.9206 ± 0.0121	0.9020 ± 0.0120

Table 3. Phase I link-prediction results, five-fold cross-validation.

This result was not reached in a single pass; Table 4 traces the three consecutive runs, on an earlier 238-model corpus revision, behind the largest gains. The biggest jump (R35→R36, +0.089 mean AUC) followed from discovering that two features — hole count and joint type — had been silently zero-valued since the project's inception; once populated and paired with a spatial link signal and a real contact-area edge weight, the encoder gained information it never actually had. The second gain (R36→R37, +0.049) came from correcting a 1:0.5 negative-sampling skew (which inflates AP independent of true ranking skill) to the standard 1:1 ratio, alongside the promotion-gate fix (Section 7). Neither gain came from architecture search or tuning; both came from fixing silently broken inputs.

Run	Change	Mean AUC
R35	Populated previously dead hole-count / joint-type features	0.539
R36	+ spatial link signal, real contact-area edge weight	0.628 (+0.089)
R37	+ neg_ratio fix, promotion-gate fix, seeded splits	0.677 (+0.049)

Table 4. Three consecutive development runs illustrating the largest single-run gains (238-model corpus revision; superseded by the 749-model results in Table 3).

9.2 Phase II: recommendation and generation

NodeRanker reaches Hit@1 of 0.4093 against a majority-class baseline of 0.3442 — a genuine improvement over always predicting the plurality class (body, 35.5% of the corpus), not the exact tie this head produced in earlier corpus revisions. Hit@5 reaches 0.9101, MRR 0.6102, NDCG@5 0.6754. Table 5 breaks Hit@1 down by type: performance is markedly uneven, body (majority) at 0.689 against washer at 0.0 and thin_plate at 0.024, and the head over-predicts body relative to its true frequency (64.7% of predictions vs. 41.3% true share) — an unresolved majority-class bias, reported honestly rather than obscured by the headline number (Section 10).

Component type	n	Hit@1	True share
body	222	0.689	34.3%
nut	36	0.444	5.6%
thick_plate	111	0.360	17.1%
bolt	98	0.316	15.1%
long_shaft	71	0.225	11.0%
short_shaft	47	0.149	7.3%
thin_plate	42	0.024	6.5%
washer	18	0.000	2.8%

Table 5. NodeRanker per-class Hit@1 on the test split.

HybridShapeGenerator reaches a test IoU of 0.6459 and Chamfer distance of 0.0377 (selected checkpoint: validation IoU 0.6640, Chamfer 0.0359), comfortably clearing target thresholds of IoU ≥ 0.35 and Chamfer ≤ 0.08. No test-set sample required the VAE fallback below the retrieval confidence threshold in this evaluation run, indicating retrieval alone currently covers the test distribution at the configured threshold — a result specific to this corpus's category coverage, not evidence the VAE component is unnecessary in general.

9.3 An engineering finding from exploratory data analysis

A full exploratory pass over the final 648-graph corpus, run after training had already completed successfully, surfaced an issue invisible to the training metrics themselves: the SDF-mean node feature (Section 3.1) is exactly zero for 25,142 of 25,155 bodies (99.95%); only 13 bodies carry a non-zero value, and the paired SDF-variance feature moves in near-perfect lockstep ($r > 0.999$) — consistent with a silently constant feature rather than genuinely near-zero geometry. This mirrors an earlier instance where hole-count and joint-type features were found silently zero-valued since the project's inception; fixing that produced the largest single-run gain of the project (Table 4). We report the SDF finding unresolved rather than omit it: strong aggregate metrics are not sufficient evidence every input feature is contributing, and targeted corpus auditing remains necessary even after training succeeds.

A second, related finding from the same analysis: 72.1% of bolts and 83.5% of washers in the corpus are flagged by the geometric hole detector as has_holes, a

higher rate than thick_plate bodies (47.8%), which structurally host the through-holes fasteners actually sit in. A washer's own bore is a genuine hole, so a high rate there is expected; a solid bolt normally is not, and the elevated rate plausibly reflects the detector picking up thread-groove or head-recess geometry as false positives on a meaningful fraction of bolts. We flag this as a candidate data-quality issue for future correction rather than a confirmed bug, since we have not yet manually verified it against individual STEP files at scale.

10 Limitations and Future Work

Three limitations are worth stating plainly. First, NodeRanker's majority-class bias (Section 9.2) means Hit@1, while a genuine improvement over baseline, is unevenly distributed across types; class-balanced sampling or loss re-weighting is the natural next step. Second, the conditional-VAE fallback is currently scoped to fasteners only; broadening it to plates and shafts is future work, alongside re-running the retrieval-threshold analysis, since the current zero-fallback result (Section 9.2) may not hold once non-fastener types are added. Third, the SDF and hole-detection findings (Section 9.3) are open issues: repairing the SDF ray-cast and auditing the bolt/washer false-positive rate against ground-truth geometry are near-term priorities before the next retrain. More broadly, the corpus is domain-specific to fastening and flow-control mechanisms; extending coverage to a wider range of assembly types is a natural direction for generalising these results.

11 Conclusion

We have presented an end-to-end graph neural network system for CAD assembly completion, structured as two phases sharing one frozen relational-attention encoder: Phase I detects missing components via inductive link prediction, and Phase II recommends the most likely missing component type and generates an actual placeable 3D shape for it through a retrieval-first, VAE-fallback generator. On a curated 749-model, seven-category industrial-parts corpus, the system reaches a mean link-prediction AUC-ROC of 0.9206, a next-component ranking Hit@1 genuinely above a majority-class baseline, and a shape-generation IoU of 0.6459 with Chamfer distance 0.0377, alongside two supporting contributions — an Octree-based open-surface detector and a component-type frequency prior effective even without graph context — and a live, natural-language explanation layer built on GNNExplainer. We have also reported, rather than suppressed, two engineering findings surfaced only through direct exploratory analysis of the training corpus after the model itself had already trained successfully, as a concrete argument that dataset auditing and model evaluation are complementary practices, neither sufficient alone.

Reproducibility. All five-fold cross-validation splits are seeded per fold and per graph, so a checkpoint's reported metrics are exactly reproducible from the same corpus snapshot. The 749-model corpus, all training and evaluation scripts, the trained checkpoints referenced in Tables 3–5, and the exploratory-analysis notebook behind Section 9.3's findings are maintained in the author's project repository and available on request.

Acknowledgments. I thank Prof. Sagarika Borah (Phase 1 guide) and Prof. Gaurav Siwal (Phase 2 guide), and the Department of Data Science and Artificial Intelligence at PES University, for guidance and review throughout this M.Tech project.

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